

Corporate Finance

Lecture 3: The Efficient Frontier and the Best Portfolio

Camilo Saldarriaga

Finance Department, ESCP Business School, 2025/2026

Plan

- ▶ Find the efficient portfolios built from two stocks
- ▶ See what short sales add to the picture
- ▶ Move from two stocks to many: the efficient frontier
- ▶ Add a risk-free asset and identify the single best risky portfolio
- ▶ Separate the risk that diversification removes from the risk that stays

Outline

- A. Efficient portfolios with two stocks
- B. N risky stocks: the efficient frontier
- C. The “super-efficient” portfolio
- ▶ The content of this lecture mostly draws on the BDM textbook Chap. 11.2-5.

Recap: a portfolio is a set of weights

- ▶ A portfolio is described by how your money is split between the stocks. Those fractions are the **weights**, and they add up to 1.
- ▶ With EUR 30 of every EUR 100 in stock 1 and EUR 70 in stock 2, the weights are $x_1 = 0.30$ and $x_2 = 0.70$, and the portfolio return is

$$R_p = x_1 R_1 + x_2 R_2$$

- ▶ The return of the portfolio is simply each stock's return counted in proportion to the money behind it.

Recap: portfolio risk depends on comovement

- ▶ Risk does not add up the same way. The variance of a two-stock portfolio is

$$\text{Var}(R_p) = \underbrace{x_1^2 \text{Var}(R_1) + x_2^2 \text{Var}(R_2)}_{\text{individual risks}} + \underbrace{2x_1 x_2 \text{Cov}(R_1, R_2)}_{\text{comovement}}$$

- ▶ The last term is what makes a portfolio different from the sum of its parts. It is positive when the two stocks tend to move together and negative when they move in opposite directions.
- ▶ The volatility of the portfolio is $\sigma_{R_p} = \sqrt{\text{Var}(R_p)}$.

Recap: correlation drives diversification

σ_p is the volatility of the portfolio, and ρ_{12} the correlation between the two stocks.

$\rho_{12} = 1 \Rightarrow \sigma_p = x_1\sigma_1 + x_2\sigma_2$ The risk of the portfolio is the average risk.

$\rho_{12} < 1 \Rightarrow \sigma_p < x_1\sigma_1 + x_2\sigma_2$ The risk of the portfolio is smaller than the average risk due to the effect of diversification.

$\rho_{12} = -1 \Rightarrow \sigma_p = |x_1\sigma_1 - x_2\sigma_2|$ The negative comovement of the assets reduces the risk of the portfolio.

A. Efficient portfolios with two stocks

Stock	Expected return	Volatility
Intel	26%	50%
Coca-Cola	6%	25%

- ▶ We assume that the correlation between the two stocks is zero.
- ▶ Which portfolio should we choose?

Efficient portfolios with two stocks (con't 2)

- ▶ Which portfolio should we choose?
- ▶ Example: portfolio invested to 40% in Intel and 60% in Coca-Cola.

Efficient portfolios with two stocks (con't 2)

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$$E[R_{40-60}] = x_I E[R_I] + x_C E[R_C] = 0.4 \times 0.26 + 0.6 \times 0.06 = 0.14$$

$$\begin{aligned}\text{Var}[R_{40-60}] &= x_I^2 \text{Var}[R_I] + x_C^2 \text{Var}[R_C] + 2x_I x_C \text{Corr}(R_I, R_C) \sigma_I \sigma_C \\ &= 0.4^2 \times 0.5^2 + 0.6^2 \times 0.25^2 + 0 \\ &= 0.0625\end{aligned}$$

$$\sigma_{40-60} = \sqrt{0.0625} = 25\%$$

Efficient portfolios with two stocks (con't 3)

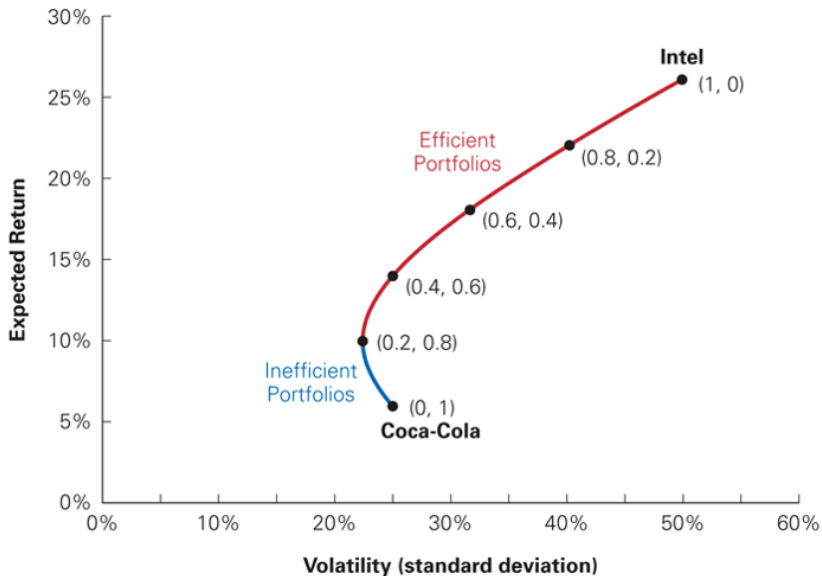
Portfolio Weights		Expected Return (%)	Volatility (%)
x_I	x_C	$E[R_p]$	$SD[R_p]$
1.00	0.00	26.0	50.0
0.80	0.20	22.0	40.3
0.60	0.40	18.0	31.6
0.40	0.60	14.0	25.0
0.20	0.80	10.0	22.4
0.00	1.00	6.0	25.0

Expected returns and volatility for different portfolios of two stocks.

But wait... “an efficient portfolio”?

- ▶ We define as efficient, all portfolios with:
 - ▶ Minimum variance for a given expected return;
 - ▶ Maximum expected return for a given variance.
- ▶ A portfolio is inefficient when it is possible to find another portfolio which has a larger expected return and a lower or equal volatility.
- ▶ Hence: investors who simultaneously maximise the expected return on their portfolio and minimise risk, will choose an efficient portfolio.

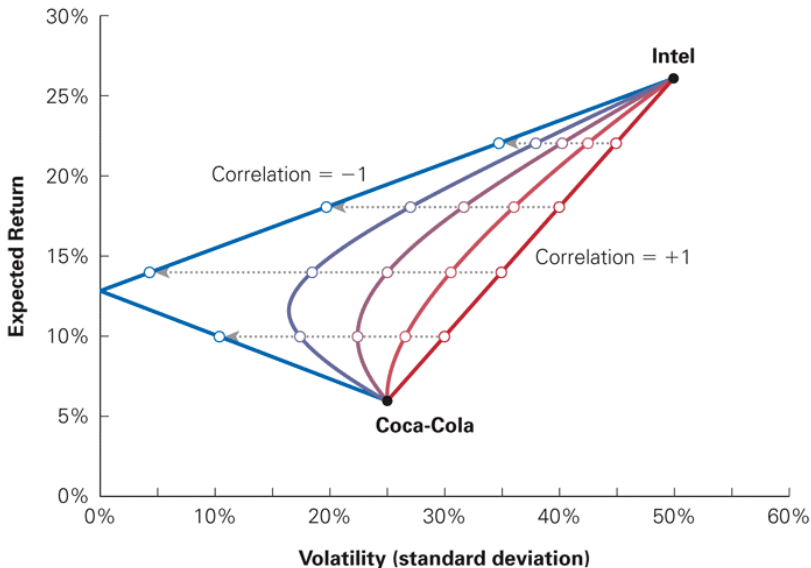
Back to our example



Improving returns with an efficient portfolio: the effect of correlation

- ▶ Consider a portfolio which consists to 100% of Coca-Cola stocks
- ▶ How can you maximize the expected return of this portfolio without increasing its volatility?

Improving returns with an efficient portfolio: the effect of correlation (con't)



The minimum variance portfolio

- It is easy to find the minimum variance portfolio in the case of two stocks:

$$\min_x \sigma_p^2 = x^2 \sigma_1^2 + (1-x)^2 \sigma_2^2 + 2(1-x)x \sigma_1 \sigma_2 \rho_{1,2}$$

The first-order condition is

$$2x\sigma_1^2 - 2(1-x)\sigma_2^2 + 2\sigma_1\sigma_2\rho_{1,2} - 4x\sigma_1\sigma_2\rho_{1,2} = 0$$

$$\Leftrightarrow x = \frac{\sigma_2^2 - \sigma_1\sigma_2\rho_{1,2}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho_{1,2}}.$$

Short sales

- ▶ Short sales make it possible to benefit from falling stock prices.
→ Strategy:
 - ▶ Take a short position: borrow and sell a stock.
 - ▶ Close a short position: buy the stock back and return it.
- ▶ Usually done when you expect the price to go down.

Short sales



Space Exploration Technologies Corp

NASDAQ: SPCX

Short sales

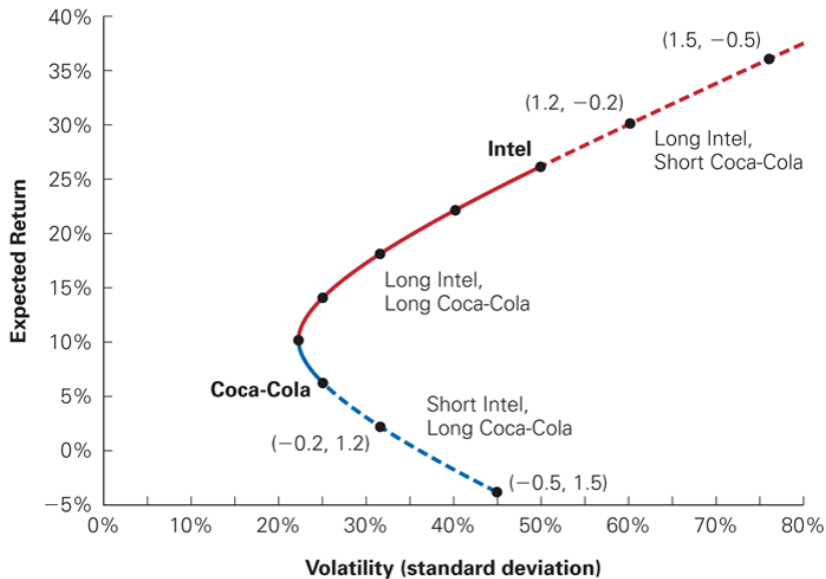
- ▶ Example: short sale of Stock A for a value of 100 and investment of the proceeds at the risk-free rate of 5% for one year. Consider the following scenarios at the end of the year:

Return of Stock A	10%	0%	-10%	-20%
Position in risk-free asset	105	105	105	105
Short position	-110	-100	-90	-80
Gain/loss	-5	5	15	25

- ▶ Mathematically, a short position corresponds to a negative portfolio weight in the stock.

Return of a portfolio with short sales

- ▶ Jérôme has 20 000 EUR available to invest for one year.
- ▶ He decides to sell short for a total of 10 000 EUR Coca-Cola stocks to buy Intel stocks for a total of 30 000 EUR
 - ▶ Remark: we still assume that the correlation between the two stocks is zero.
- ▶ What is the expected return and the volatility of his portfolio?

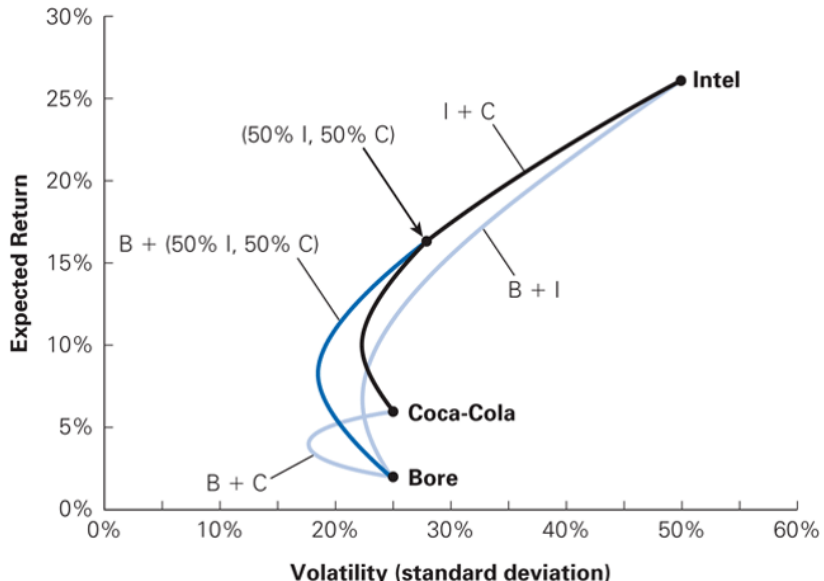


B. N risky stocks: the efficient frontier

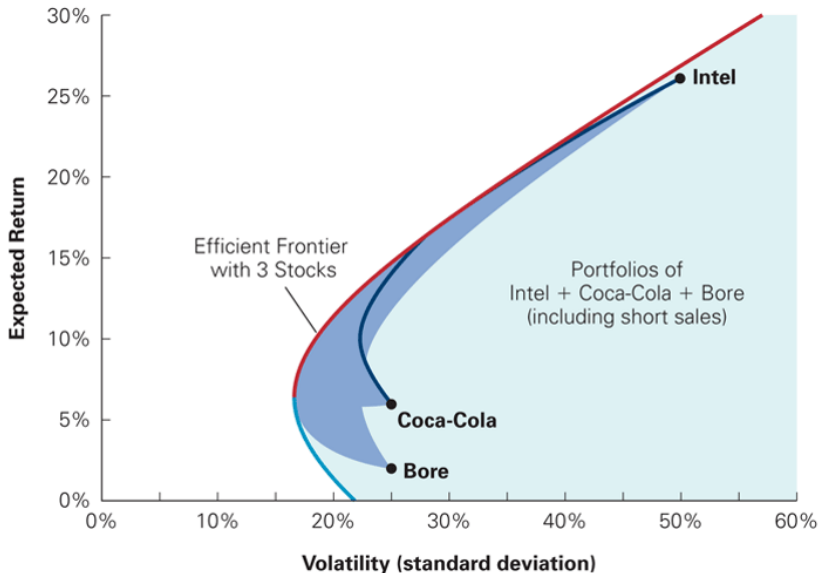
- ▶ What is the effect of adding a third stock to a portfolio of Coca-Cola and Intel?
- ▶ We assume that the stock of Bore Industries is uncorrelated with the other two stocks and that its expected return is only 2% and its volatility is the same as the one of the Coca-Cola stock (25%).
- ▶ Reminder: You would have to use this formula:

$$\begin{aligned}\text{Var} [R_p] &= \text{Cov} [R_p, R_p] = \sum_{i=1}^N x_i \text{Cov} [R_i, R_p], \\ &= \sum_{i=1}^N x_i \text{Cov} \left[R_i, \sum_{j=1}^N x_j R_j \right] = \sum_{i=1}^N x_i \sum_{j=1}^N x_j \text{Cov} [R_i, R_j] \\ &= \sum_{i=1}^N \sum_{j=1}^N x_i x_j \text{Cov} [R_i, R_j].\end{aligned}$$

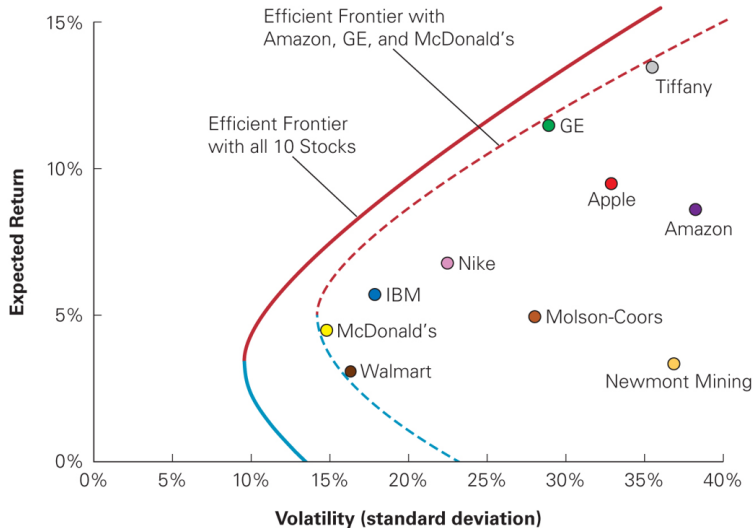
Expected return and volatility for selected portfolios of Intel, Coca-Cola, and Bore Industries stocks



Volatility and expected return for all portfolios of Intel, Coca-Cola, and Bore stock



Efficient frontier with three versus ten stocks



Summary on efficient portfolios

- ▶ We define as efficient, any portfolio with minimum variance for a given expected return.
 - ▶ Or: any portfolio with maximum return for a given level of risk.
- ▶ The efficient frontier is then the set of efficient portfolios.
 - ▶ It can be shown that it is the upper part of a hyperbola in the $(E(R), \sigma)$ plane.
- ▶ Augmenting the number of securities included in a portfolio allows to obtain an efficient portfolio with a lower risk, for a given level of expected return.
- ▶ Remark: an efficient portfolio may contain some short positions.

C. The “super-efficient” portfolio

- ▶ Up to now, the efficient portfolio only contained risky securities.
 - ▶ The investment decision depends on the preferences (“risk vs return”) of each investor
- ▶ Let us allow investors to invest in a risk-free asset
 - ▶ Its return is denoted by r_f (risk-free rate)
 - ▶ Zero risk

Risky asset and risk-free investment: the effect of leverage (James Tobin, 1958)

Two investments are available:

► Stocks of Umbrella Inc:

Return		Expected return	St.Dev.
Sun 1-2 -10%	Rain 30%	10%	20%

► A risk-free bank deposit

Return		Expected return	St.Dev.
Sun 1-2 3%	Rain 3%	3%	0%

Risky asset and risk-free investment: the effect of leverage (con't)

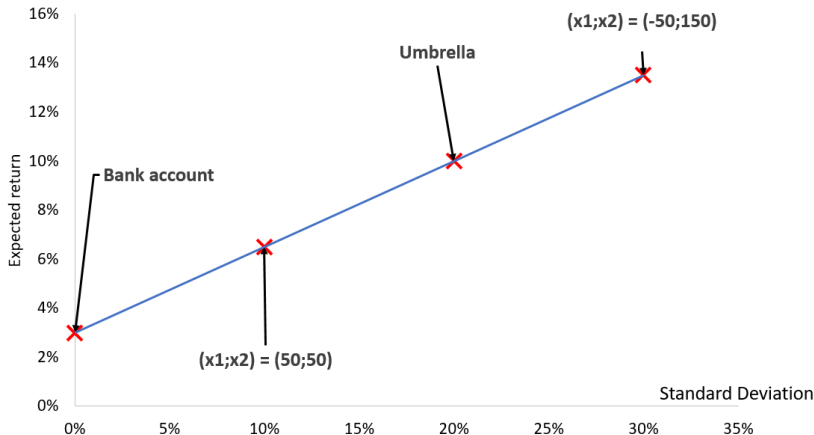
Invest 50 in Umbrella Inc and 50 in the bank account:

Risky asset and risk-free investment: the effect of leverage (con't 2)

Borrow 50 and invest 150 in Umbrella Inc:

Risky asset and risk-free investment: the effect of leverage (con't 3)

Possible combinations in the risk-return diagram



Leverage effect: general formula for the expected portfolio return and risk

- Expected return: weighted average (by the fraction invested) of each constituent's return:

$$E(R_p) = \sum_{i=1}^N x_i \times E(R_i) + x_f r_f, \text{ where assets } i = 1, \dots, N \text{ are risky and } x_f + \sum_{i=1}^N x_i = 1.$$

- Risk:

If $N = 1$, then $\sigma_p = x\sigma$, where x is the fraction invested in the risky asset and σ is its volatility.

If $N > 1$, then it suffices to calculate portfolio variance using only the risky assets with

$$\sigma_p^2 = \sum_{i=1}^N x_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N x_i x_j \text{Cov}(R_i, R_j),$$

and take $\sigma_p = \sqrt{\sigma_p^2}$.

Leverage effect: consequences

The possibility to change the risk and return of a portfolio with a risk-free security has two fundamental consequences.

1. It is easy to generate high returns ...if a high level of risk is taken. $\Rightarrow$ Never analyse a return without also analysing its risk.
2. It is now possible to select objectively between two risky investments: $\Rightarrow$ The portfolio which generates the highest slope in the $(E(R), \sigma)$ plane is preferred, **regardless of the risk aversion**.

Sharpe ratio: identification of the best investment

- ▶ Sharpe ratio:

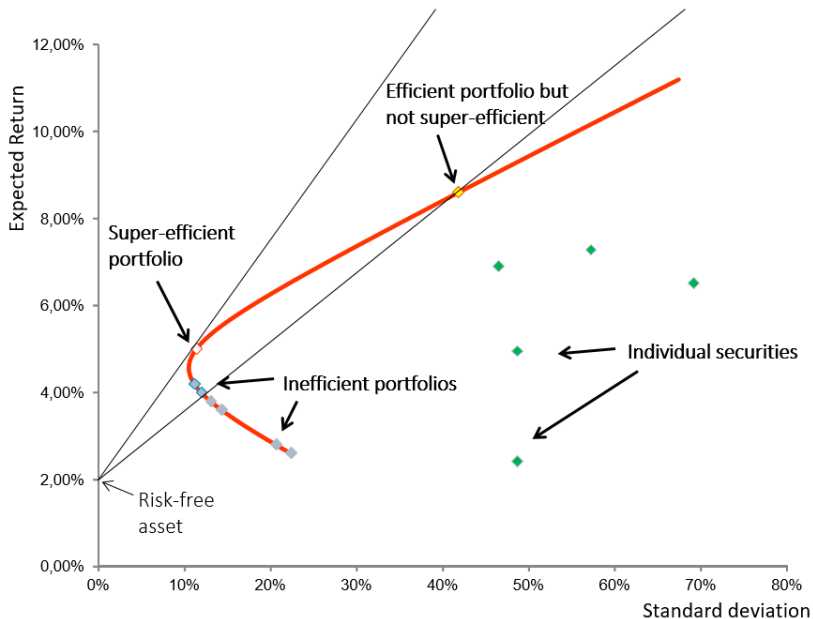
$$\frac{\text{Excess return of the portfolio}}{\text{Portfolio volatility}} = \frac{E[R_p] - r_f}{\sigma_p}.$$

- ▶ **Remark:** Here, the best investment does not depend on the aversion to risk. $\Rightarrow$ Regardless of the risk aversion, all investors should hold the portfolio of risky assets

What is the link between leverage, diversification, and efficient portfolios?

- ▶ If investors can invest and borrow at the risk-free rate, only one risky portfolio is efficient: the “tangency portfolio” or “super-efficient portfolio.”
- ▶ Conclusion:
 - ▶ From a given set of risky securities, investors should construct the same portfolio.
 - ▶ They will then adjust for their risk-aversion by investing/borrowing at the risk-free rate. $\Rightarrow$ Two fund separation theorem (James Tobin)
- ▶ It can then be shown that the efficient frontier is a half straight line, tangent to the Markowitz's hyperbola. The tangency point is the ...tangency portfolio.

The efficient frontier with a risk-free asset



Summary on efficient frontier with a risk-free asset

- It can be shown that the equation of the efficient frontier WITH risk-free asset (e.g., with borrowing/leverage) is given by:

$$E(R_p) = r_f + \underbrace{\frac{E(R_T) - r_f}{\sigma_T}}_{\text{Risk premium}} \sigma_p,$$

where the subscript p corresponds to the efficient portfolio and the subscript T to the tangency (or “super-efficient”) portfolio

- **Reminder:** all efficient portfolios contain the tangency portfolio and the risk-free asset (long or short)

Two types of risk

1. **Systematic risk**: common to all securities.



2. **Idiosyncratic risk**: specific to one security.

Diversification and risk

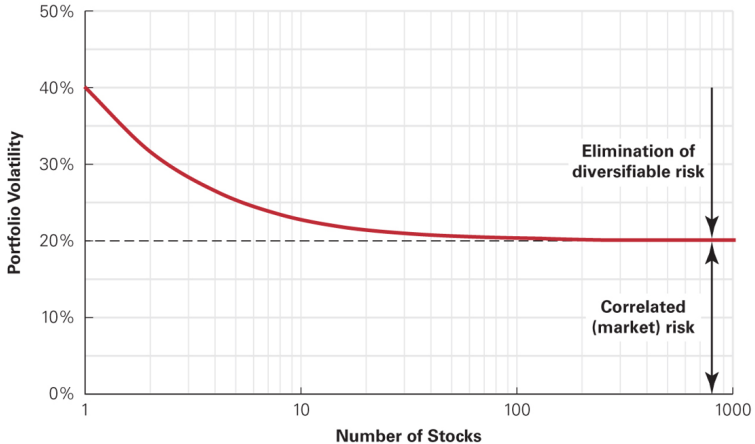
- ▶ Which risk do investors reduce by diversifying their portfolios?
 - ▶ The **systematic** risk? [risk common to all securities]
 - ▶ The **idiosyncratic** risk? [risk specific to a particular security]
- ▶ Remark: the variance of any portfolio can be written as

$$\sigma_p^2 = \sum_{i=1}^N x_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N x_i x_j \text{Cov} [R_i, R_j] .$$

- ▶ Not clear? Take $x_i = 1/N$ (equally-weighted portfolio)

Diversification and risk: a simple example

Volatility of an equally-weighted portfolio versus the number of stocks



Diversification and risk: summary

Common risk Market-wide:
economic conditions, interest
and exchange rates, inflation, ...
**The asset is affected by the
evolution of the market** (with
its own sensitivity)



Systematic risk =
Undiversifiable risk (market
risk)

Independent risk Specific to the
firm/security i : product market,
quality of the management,
operational risk, ...



Idiosyncratic risk =
Diversifiable risk

Exercise 1: Optimal Portfolio Choice with one risky asset

- ▶ Stock A trades today for a price of EUR 100. Depending on the evolution of the economy, the price of Stock A in one year will be:
 - ▶ 120 with probability 0.5 (expansion)
 - ▶ 100 with probability 0.5 (recession)
- ▶ The risk-free rate is 5%
- ▶ The investment fund AlwaysMore claims to offer an investment product which is superior to Stock A. For an initial investment of 220, it commits itself to pay in one year:
 - ▶ 320 in the expansion
 - ▶ 200 in the recession.
- ▶ Should you accept this offer? If not, what is the highest price you are willing to pay for this investment?

Exercise 2: Optimal Portfolio Choice with two risky assets

- You want to form a portfolio with two stocks A and B. These stocks have expected returns of $E_A = 10\%$ and $E_B = 15\%$ with standard deviations of $\sigma_A = 15\%$ and $\sigma_B = 25\%$. The correlation coefficient between the two stocks is -0.3 .
1. What is the risk and return of a portfolio invested to 40% in A and 60% in B?
 2. If you want to achieve an expected return of 11%, how much risk do you need to take?
 3. Suppose that you can invest in the two stocks and also invest and borrow at the risk-free rate of 5%. Show that you can construct a portfolio with the same return than stock A but with a lower risk.

Exercise (diversification)

- ▶ You can invest in two stocks with the following characteristics

Stock	Expected Return	St.Dev.
A	12%	15%
B	20%	45%

- ▶ What is the risk and the return of a portfolio invested to 50% in A and 50% in B if the correlation coefficient is 0.2?
- ▶ If the stocks are perfectly negatively correlated, what is the portfolio with the lowest risk?

Quiz

- ▶ Define the leverage effect
- ▶ Define the diversification effect
- ▶ Is it possible to augment the return of a portfolio with the risk-free asset?
 - ▶ What happens then to the risk of the portfolio?
- ▶ If the stocks are not perfectly correlated, then the risk of the portfolio is always: a) lower, b) larger c) equal to the weighted average of the individual risks?

Quiz

- ▶ Is it possible to fix the level of risk of a portfolio and how?
- ▶ Is it possible to fix the level of expected return of a portfolio and how?
- ▶ In which case can it be advantageous to buy a stock with higher risk and lower expected return?
- ▶ What are short sales?
- ▶ What are you betting on in the case of a short sale?
- ▶ What does a short position correspond to in terms of portfolio weight?